

Momentum & Reversal Engine

Period analysis · 2024-09-01 to 2026-09-17 · 102 weekly scans · generated 17 Sep 2026 15:21 IST

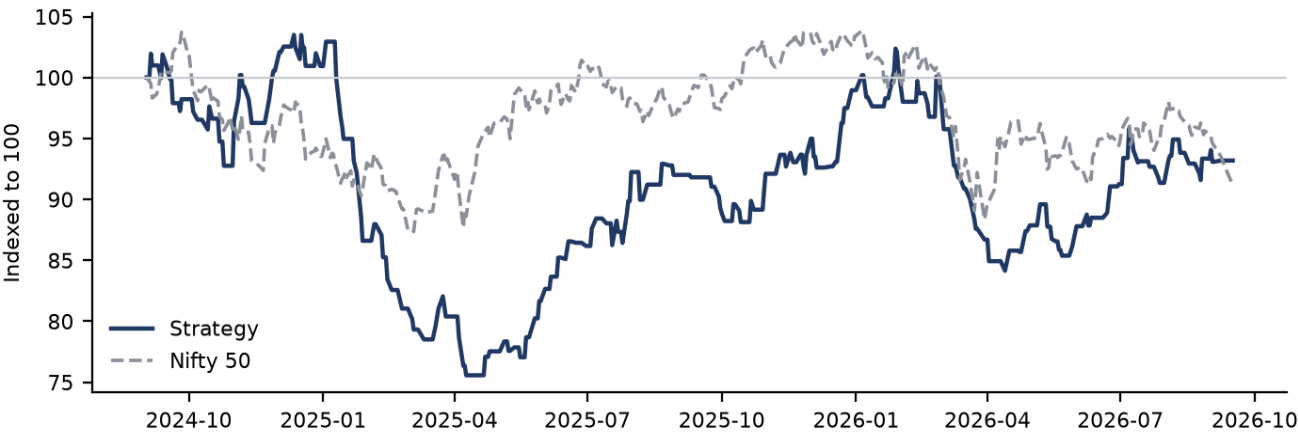
The strategy lost money over these two years and beat the index only because the index fell further. Alpha of +0.8% over 2.04 years is inside noise, and it was earned with a 27% drawdown.

Headline

Metric	Result	Benchmark
Starting capital	1,000,000	-
Ending equity	931,844	-
Net P&L	-68,156	-
Total return	-6.8%	-7.6%
CAGR	-3.4%	-3.8%
Alpha (total / CAGR)	+0.8% / +0.4%	-
Max drawdown	-27.0%	-
Capital deployed	37,954,820	37.95x turnover
Positions taken	264	955 skipped, no cash

Sizing: risk 1% of equity per trade across the entry-to-stop distance, at most 10 open positions and 20% of equity in any one name; entry at the next session's open; 0.25% round-trip costs. Percentage statistics below are sizing-independent and hold at any capital.

Equity curve



955 signals were skipped for lack of cash against 264 taken. The screen generates roughly five times more positions than a 10 lakh account can fund, so results at this capital reflect the subset that fitted, not the full signal set.

1. Trade statistics

All 1,362 momentum and reversal trades in the window. These are signal-level and do not depend on position size.

Measure	Value	Measure	Value
Trades	1,362	Average hold	10.52 days
Win rate	37.2%	Median hold	9 days
Average win	+8.3%	Hold, winners	13.64 days
Average loss	-5.1%	Hold, losers	8.68 days
Expectancy	-0.09%	Average R	-0.011
Profit factor	0.972	Realised reward:risk	1.65 : 1
Longest win streak	10	Longest loss streak	18

Best and worst

	Symbol	Date	Return	R	Held	Exit
Best	CEMPRO	2026-06-08	+33.9%	1.19	20d	time
Worst	ITI	2025-01-03	-34.5%	-1.0	17d	stop

The best trade exited on time at only 1.19R, not at target - the 2R target was never reached despite a 34% gain, because the move came after the 20-session clock ran out. The worst was a reversal trade whose stop sat 34% below entry: reversal stops anchor to the 15-session low, which can be far away. It lost exactly 1R as designed, but 1R was a third of the position.

Exit mix

Exit reason	Share of trades
stop	53.8%
target	23.3%
time	22.9%

By model

Model	Trades	Win rate	Expectancy	Profit factor	Avg R
control_naive_low	846	36.8%	-0.01%	0.995	-0.011
momentum	1,219	35.7%	-0.23%	0.927	-0.029
reversal	143	50.3%	+1.13%	1.345	0.146

Momentum produced 1,219 trades at a profit factor of 0.93 - it lost money. Reversal produced 143 at 1.35 and made money. The engine spends almost all of its activity on the model that does not work.

2. Where the money was lost

2.1 A data error, not a trading error

The single worst trade in the first run was Vedanta at **-60.3%**, a -10R loss. It never happened. On 30 April 2026 VEDL gapped from 773.60 to 289.50 overnight: the demerger moved value to separately-listed entities. Yahoo's adjustment handles splits and dividends, not demergers, so the price series shows a collapse the shareholder never took.

Eight such discontinuities were found across the universe (VEDL, ABFRL, TMPV, TRENT, GPIL, CGCL, MOTILALOFS, PATANJALI). The engine now detects them - an overnight gap worse than -30% that never retraces - and excludes any trade spanning one, rather than inventing adjusted prices for value that left the listing. Every figure in this report is computed after that fix.

2.2 Stops

Measure	Value
Trades stopped out	733 (53.8% of all)
Would have reached target without the stop	97 (13.2%)
Closed above entry by the 20-day horizon	207 (28.2%)
Average loss when stopped	-5.4%

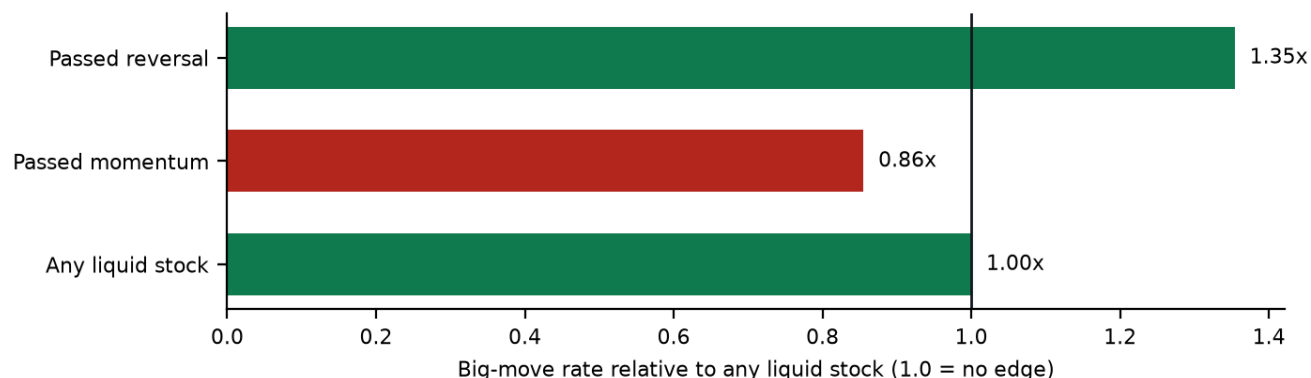
28% of stopped trades finished above entry within the horizon, and 13% would have hit target outright. That is the measurable cost of the stop distance. It is not proof the stop is wrong: it exists to cap the other 72%, whose average loss was -5.4%. Widening it would recover some of those 207 trades and deepen the rest.

2.3 The loss streak

The longest run of consecutive losers was **18**, against a best winning run of 10. At 37% win rate a run of 18 is not anomalous - it is what a 37% process does regularly. Any position sizing that cannot survive 18 straight losses is mis-specified, independent of the edge.

3. What the filters missed

Across the window, 2,755 stock-dates in the liquid universe produced a gain of 15% or more within 20 sessions. The momentum screen surfaced 426 of them - a capture rate of **15.5%**.



This is the finding that matters most. A stock picked at random from the liquid universe had a 6.69% chance of a +15% move. One that passed every momentum rule had 5.72% - **a lift of 0.86x, meaning the momentum filter selected stocks slightly less likely to run than chance**. Reversal reached 9.06%, a genuine 1.35x edge.

Which rule rejected the winners

Rule	Blocked alone	Big movers blocked	Hit rate of blocked	vs baseline
ma_aligned	2,705	124	4.6%	0.68x
near_52w_high	830	75	9.0%	1.35x
not_extended_50dma	352	36	10.2%	1.53x
rs_positive	1,036	30	2.9%	0.43x
not_extended_atr	318	15	4.7%	0.71x
atr_floor	75	2	2.7%	0.40x
atr_ceiling	15	1	6.7%	1.00x

Read the last column, not the third. **ma_aligned** blocked the most big movers in absolute terms, but the pool it rejected had a below-baseline hit rate - it is working. **not_extended_50dma** and **near_52w_high** rejected pools that beat the baseline, which is the signature of a filter throwing away good stocks.

The largest individual misses

Date	Symbol	20-day gain	Rules failed
2026-06-08	AEGISLOG	+82.2%	near_52w_high, ma_aligned
2025-05-12	GRSE	+78.3%	near_52w_high, atr_ceiling, ma_aligned
2025-04-25	GRSE	+67.2%	near_52w_high, rs_positive, ma_aligned
2026-04-08	CEMPRO	+64.8%	near_52w_high, rs_positive, ma_aligned
2026-07-07	KALYANKJIL	+64.3%	near_52w_high, rs_positive, ma_aligned
2025-06-23	GABRIEL	+63.3%	rs_positive, ma_aligned
2025-09-03	NETWEB	+60.0%	near_52w_high, not_extended_50dma, ma_aligned
2024-12-05	ITI	+59.8%	near_52w_high, rs_positive, ma_aligned
2025-04-17	APARINDS	+58.0%	near_52w_high, rs_positive, ma_aligned
2026-04-08	COHANCE	+57.8%	near_52w_high, rs_positive, ma_aligned

Almost every large miss failed **ma_aligned** and **near_52w_high** together: these were stocks not yet in an established uptrend and not near their highs - early-stage recoveries. A momentum screen is designed not to buy those. They are the reversal model's territory, and reversal is the model that works.

4. Acting on the misses makes it worse

The obvious response to section 3 is to relax the two filters that reject good stocks. Each variant below was judged on the second year, which never informed it. A change has to improve both halves to be worth keeping.

Variant	IS trades	IS profit factor	OOS trades	OOS profit factor	Verdict
baseline	613	0.842	606	1.033	reference
relax_ext_50dma	615	0.83	618	0.976	reject
relax_52w_high	618	0.811	605	1.046	reject
relax_both	620	0.813	618	0.987	reject
drop_ext_50dma	615	0.832	618	0.987	reject
tighten_rs	613	0.826	602	1.044	reject

Every relaxation was rejected. The miss analysis counts only the winners a filter blocked; it is silent on the losers it blocked at the same time. Relaxing admits both, and the losers dominate. Note also that trade counts barely move - 613 to 615 - because the score ranking already demotes those stocks, so the filter rarely changes the final twenty. A rule can look expensive at universe level and be nearly irrelevant to what actually gets bought.

No thresholds were changed as a result of this analysis. That is the correct outcome, not a failure to act.

5. What to do

1. Stop trading the momentum model at full size

1,219 trades at a 0.93 profit factor and a 0.86x selection lift. Two years is enough evidence to say it is not working as specified, and it consumes 90% of the activity.

2. Investigate widening the reversal model instead

143 trades, 1.35 profit factor, 1.35x lift, and it held up in both halves of the full backtest. Its weakness is sample size, not quality. Loosening reversal's entry conditions is the one change with an evidential basis - and it must clear the same out-of-sample gate.

3. Cap reversal stop distance

Reversal stops anchor to the 15-session low and reached 34% of entry. A 1R loss is survivable; a 1R loss that is a third of the position is a sizing problem. Cap risk per trade at a fixed fraction of price and skip signals that exceed it.

4. Keep the corporate-action guard and extend it

One unadjusted demerger created a phantom -10R loss. Eight were found. Any vendor feed will keep producing these; the detector should run on every refresh and the flagged list reviewed.

5. Do not tune on this report

Section 4 is the demonstration. Every change suggested by the miss data failed validation. Treat section 3 as a description of the strategy's blind spot, not a to-do list.

Limits of this report. The universe is today's Nifty 500, so companies that dropped out or delisted are absent and every figure is optimistic. Fills are assumed at the next open with no slippage beyond the cost assumption, no circuit limits and no impact. When a stop and target fall on the same day the stop is assumed. The window covers a falling market - the Nifty lost 7.7% - and momentum strategies are known to struggle in those conditions, so this is not a verdict on the approach across all regimes. Research output, not investment advice.